

CiteView: In-Situ Citation Sensemaking for Scholarly Reading

ANONYMOUS AUTHOR(S)

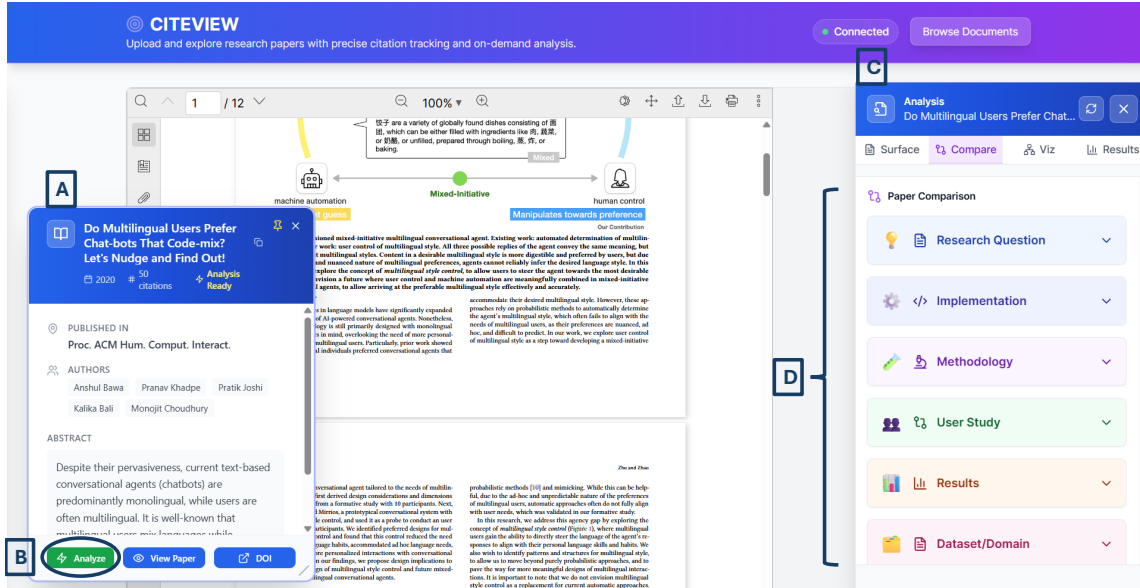


Fig. 1. CiteView interface overview. (A) Paper metadata card displays key bibliographic information. (B) The *Analyze* button initiates on-demand processing of the paper and its citations, triggering extraction of citation contexts, retrieval of cited papers, and generation of multi-modal analysis. (C) The Analysis sidebar opens when users click citations, presenting four complementary tabs: Surface, Compare, Viz, and Results. that respectively surface the most relevant context and citation reasons, contrast the citing and cited papers, visualize the cited work through figures and a semantic graph of its flow, and foreground its main empirical findings.(D) The Compare tab organizes side-by-side comparisons across six research dimensions, including Research Question, Implementation, Methodology, User Study, Results, and Dataset/Domain, each expandable to reveal detailed paired descriptions.

Academic papers contain extensive citations, yet readers often lack the context needed to understand why and how cited works are relevant. We present CiteView, an AI-assisted citation analysis tool that turns inline citations into entry points for contextual understanding. Clicking a citation opens a compact sidebar with layered, citation-aware insights. The sidebar presents contextualized passages from cited papers, citation intent, distilled key results, concise comparisons, and visual links between local argument flow and referenced figures. These capabilities are supported by a multi-agent pipeline for extraction, retrieval, relevance ranking, and synthesis, enabling fast and traceable responses. We evaluated CiteView through a mixed-methods laboratory study with (n=12) participants performing citation-focused reading tasks. We collected interaction logs, post-task questionnaires (SUS and NASA-TLX), and semi-structured interviews. Participants reported positive usability, low workload, and high efficiency. Qualitative feedback

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indicated reduced friction in accessing cited-paper contributions and a preference for insight-driven views focused on highlighting results, enabling direct comparisons, and supporting visual interpretation.

CCS Concepts: • **Do Not Use This Code** → **Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Additional Key Words and Phrases: Do, Not, Us, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 Introduction

Citations serve as the foundation of scholarly knowledge, linking new contributions to prior work and enabling readers to trace the evolution of ideas across papers [15, 28]. Yet understanding how a citation actually contributes to an argument remains surprisingly difficult in practice. For example, whether it provides supporting evidence, methodological precedent, or contrasting findings,. Readers encountering a citation face a disruptive choice: accept the author’s characterization of the cited work on faith, or break their reading flow to manually verify the claim by navigating to the source paper [20, 26]. This context-switching imposes a significant cognitive load [33, 38] and often leads readers to either skip verification entirely, risking misinterpretation [2], or invest substantial time investigating each reference at the expense of deeper analysis [16, 26].

Despite citations being fundamental to scholarly argumentation, existing reading tools offer limited support for accessing the specific context needed to interpret them [12, 19, 20]. Existing tools address parts of this problem but not the core challenges, namely the *fragmentation* resulting from a mismatch between researchers’ need to maintain reading continuity and the insufficient support for *context-aware citation information* revealing citation intent. Text-based assistants like SciSpace Copilot [30] clarify citing passages but do not reveal how cited papers support claims. Citation exploration platforms such as VOSviewer [36] visualize co-citation and bibliographic coupling networks but cannot extract contextually relevant content from the papers themselves. Reference managers such as Zotero, Mendeley, and EndNote¹ organize bibliographic metadata but offer limited support for understanding citation intent or retrieving figures and tables from cited papers. As a result, researchers must navigate between multiple systems to access information relevant to the passage they are reading.

To address this gap, we developed CiteView (Figure 1), an AI-assisted tool that integrates *reading process* and *contextual citation analysis*, within a single interface. In the proposed system, the user can upload a paper in PDF format and view its metadata (Figure 1A). By clicking *Analyze* (Figure 1B), the user triggers the citation analysis process and *Analysis* sidebar opens (Figure 1C) with four tabs: *Surface* (context-aware explanations and citation purpose), *Compare* (cross-paper analysis), *Viz* (semantic graphs and figures), and *Results* (quantitative and qualitative findings). These tabs present complementary information about what the cited paper is about, why it was cited, and how it connects to the current argument. The Compare tab (Figure 1D) provides side-by-side analysis across six research dimensions (Research Question, Implementation, Methodology, User Study, Results, Dataset/Domain). By surfacing citation insights and figures in a contextual sidebar, CiteView enables exploration without breaking reading flow or switching between tools.

¹Zotero: <https://www.zotero.org/>; Mendeley: <https://www.mendeley.com/>; EndNote: <https://endnote.com/>

We evaluated CiteView through a mixed-methods user study with 12 participants, who completed citation-focused reading tasks using the system. Results showed high task completion (97.9% accuracy), positive usability ratings (SUS: $M=71.9$, above industry benchmarks), and low perceived workload on NASA-TLX effort dimensions (Mental Demand: $M=2.0/10$; Effort Required: $M=2.1/10$), alongside strong ratings for time efficiency ($M=8.9/10$) and ease of finding information ($M=9.1/10$). Interview feedback highlighted that CiteView substantially reduced friction compared to traditional citation workflows, with participants particularly valuing insight-driven views (Results, Compare, Visualization) that quickly revealed what cited papers contributed and how they connected to the current work. While participants noted opportunities to streamline the interface and improve visual explanations, our findings suggest that integrating contribution-level citation context directly into the reading workflow can support faster and more effective citation exploration without disrupting the reading flow or substantially increasing cognitive load.

This paper makes two primary contributions:

- (1) **CiteView, an integrated reading interface that surfaces citation context without disrupting flow.** CiteView combines textual summaries, visual evidence extraction, and cross-paper relationship analysis to provide multi-modal citation understanding directly within the reading interface, eliminating context switching between tools.
- (2) **Empirical evidence that integrated citation support improves scholarly sensemaking.** A mixed-methods user study ($n=12$) showed high task completion (97.9%), positive usability (SUS: $M=71.9$), and low perceived workload, with participants reporting substantially reduced friction and strong preference for insight-driven views over purely descriptive summaries.

Together, these contributions advance the design of intelligent reading tools that make scholarly exploration more fluid, accessible, and cognitively manageable.

2 Related Works

Our work builds upon three intersecting areas of research: interactive tools for scholarly reading and sensemaking, citation network visualization systems, and recent advances in AI-assisted document understanding. Significant progress has been made in each domain, but there remains a lack of integrated support for contextual citation analysis during active reading of scientific papers.

2.1 Tools for Scholarly Reading & Sensemaking

Prior research on scholarly reading has consistently shown that academic reading is not a purely linear activity, but instead involves iterative navigation, annotation, and cross-referencing as readers work to interpret, evaluate, and organize ideas [17]. Early electronic reading environments such as XLibris and LiquidText [31, 35] focused on supporting active reading through flexible navigation and annotation. For instance, LiquidText [35] enabled readers to extract passages and spatially reorganize them on an infinite canvas, reflecting the non-linear strategies scholars often use to synthesize ideas. These systems demonstrated that rich annotation and navigation mechanisms can help readers identify and reason about relationships within a single document.

Building on this foundation, recent tools have introduced intelligent features that augment PDFs directly to reduce reading friction. One common approach provides contextual definitions and explanations at the point of need: ScholarPhi [14] surfaces just-in-time definitions and symbol explanations where technical terms appear, Paper Plain [3] adds plain-language summaries of medical sections to guide non-expert readers, and gaze-assisted interfaces use eye-tracking to fetch definitions when readers linger on unfamiliar terms [4]. While effective for vocabulary comprehension, these tools focus on explaining what terms mean rather than revealing how cited works support the claims being made.

Another set of tools focuses on improving navigation and readability within a single paper. Scim and PaperMage [11, 20] automatically highlight important sentences and reflow paper layouts for easier reading, helping readers identify key passages and navigate dense academic prose more efficiently. However, like definition systems, their focus remains on the current paper rather than connecting readers to the cited sources that provide evidence for its claims.

Together, these systems demonstrate how contextual augmentation can make local interactions more fluid, yet they largely treat the PDF as an isolated artifact rather than an entry point into a broader network of related evidence, figures, and ideas.

Research on sensemaking emphasizes that meaningful comprehension requires synthesizing relationships between a paper’s content and external scholarly knowledge, rather than interpreting the text in isolation. Studies of literature sensemaking and analytic provenance (e.g., SensePath) show that scholars move iteratively between documents, notes, and visual representations as they connect evidence across sources [23, 24]. Despite these findings, most digital reading tools continue to treat PDFs as self-contained artifacts, offering limited support for incorporating external context—such as concrete findings, data, or visual evidence from cited works, directly into the reading view. As a result, readers must interrupt their workflow and switch contexts to verify claims or understand how an idea is supported, increasing cognitive load and disrupting sustained engagement. Some recent systems have begun to address this limitation by integrating citation context into the reading process itself. For example, PaperQuest [25] uses citation links to suggest relevant papers during literature review, though it primarily functions as a planning aid rather than an in-document assistant. More recently, the Semantic Reader platform has inspired prototypes that overlay citation-related insights directly onto PDFs: CiteSee [5] augments inline citations with personalized relevance indicators (“paper cards” summarizing abstracts, authorship, and reading history), while CiteRead [26] automatically annotates documents with in-situ commentary from subsequent works, illustrating how later research builds on the current paper without requiring readers to leave the document.

While these context integrations are promising, they primarily focus on incoming citations or personalized signals rather than deeply analyzing outgoing references or surfacing visual evidence directly tied to the cited claim. In general, even the best current tools stop short of fully connecting a PDF to the broader web of scholarly knowledge at the exact moment of reading, leaving a gap in in-situ, citation-driven comprehension support.

2.2 Citation Analysis & Visualization Systems

To help researchers navigate the exponential growth of literature, numerous systems have been developed to visualize citation networks and bibliometric patterns. Traditional tools like CiteSpace², VOSviewer³, and CitNetExplorer⁴ provide powerful bibliometric overviews, enabling users to identify clusters, trends, and seminal works within large datasets [6, 36, 37]. Similarly, interactive web-based platforms leverage citation networks for literature discovery. Tools like Connected Papers, ResearchRabbit, Inciteful, and Litmap⁵ allow researchers to start from a seed paper and explore its citation neighborhood through co-citation and bibliographic coupling relationships. By visualizing which papers are frequently cited together or share similar reference lists, these platforms help researchers quickly identify seminal works, recent developments, and thematic clusters within a research area. However, while effective for discovery and

²CiteSpace: <http://cluster.cis.drexel.edu/~cchen/citespace/>

³VOSviewer: <https://www.vosviewer.com/>

⁴CitNetExplorer: <https://www.citnetexplorer.nl/>

⁵Connected Papers: <https://www.connectedpapers.com/>; ResearchRabbit: <https://www.researchrabbit.ai/>; Inciteful: <https://inciteful.xyz/>; Litmap: <https://www.litmap.com/>

high-level landscape mapping, they operate exclusively on metadata and cannot surface the specific content, findings, or figures that make a cited paper relevant to the claim at hand.

However, these systems primarily operate at the macro-level of citation networks rather than the micro-level of individual claims. While they can show that two papers are connected, they cannot explain how specific cited content supports particular arguments or surface the relevant figures and findings that make a citation meaningful. Most tools also depend on pre-indexed databases [1, 7] and cannot analyze user-provided PDFs on demand. Research prototypes like Action Science Explorer have attempted to bridge network visualization with document content [9], but were designed for corpus-level exploration rather than supporting in-situ citation analysis within the reading flow. This gap motivates our development of CiteView, which integrates citation context directly into the reading interface.

2.3 AI-Assisted Document Understanding

Recent advances in large language models (LLMs) and natural language processing have led to AI-powered reading assistants that help researchers understand complex academic papers. Systems such as ExplainPaper [10], SciSpace (formerly Typeset) [29], and AI literature assistants including Elicit [13] and Consensus [8] use generative AI to summarize content, simplify technical language, and answer natural language questions about documents. In parallel, smart citation platforms such as scite.ai [27] classify citation statements as supporting, contrasting, or neutral, adding semantic context beyond citation counts. Together, these tools reduce barriers to comprehension and support cross-paper synthesis, but they typically treat documents as unstructured text and overlook important visual and structural elements, such as figures, tables, and precise citation locations. As a result, AI-generated outputs may lack the specificity needed to understand how a cited work supports particular claims in context.[18].

Emerging efforts such as the Semantic Reader project aim to integrate AI-driven context directly into reading interfaces by analyzing document structure and linking it with a large academic corpus [21]. Features such as inline Citation Cards reduce context switching by presenting key details and TL;DR summaries at the point of citation, while related AI-augmented readers like CiteRead and CiteSee provide in-situ annotations based on follow-on commentary or personal reading history [5, 26]. Despite this progress, these systems rely on indexed corpora, cannot process arbitrary user-uploaded PDFs, and offer limited support for extracting figure-level evidence or enabling comprehensive, on-demand verification of cited content. As a result, AI reading assistants primarily facilitate textual understanding at the document level and lack integrated multimodal support for citation intent and visualized evidence within the native reading context [21, 27]. *CiteView* addresses this gap by combining in-context citation analysis, figure extraction, and cross-paper relationship reasoning within an interactive PDF interface, enabling seamless verification and exploration of scholarly claims without disrupting the reader’s flow.

In summary, prior work has advanced important components of the scholarly workflow: active reading tools support annotation, citation network systems enable discovery, and AI-driven services provide summarization and semantic insight [6, 21, 22, 27, 32, 35–37]. However, these capabilities remain fragmented. Reading tools rarely integrate external sources, visualization platforms operate away from the moment of reading, and AI assistants function outside the natural flow of PDF-based engagement [21, 27]. Even systems that incorporate limited context—such as CiteSee or CiteRead do not deliver concrete evidence from cited papers directly into the reading view [5, 26].

This fragmentation motivates the need for an integrated, in-document approach. *CiteView* addresses this gap by treating the PDF as an entry point to a connected scholarly knowledge graph rather than a static endpoint. Unlike tools that rely on context switching or pre-indexed databases [6, 21, 27], *CiteView* operates on-demand on arbitrary PDFs to surface citation intent, comparative relationships, and visual evidence from referenced figures or tables directly

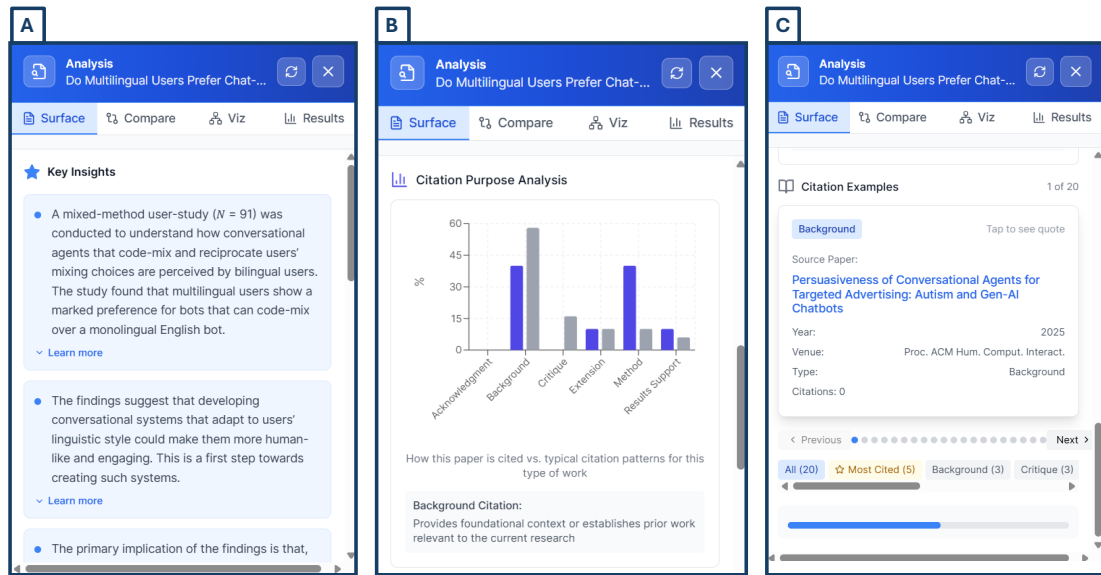


Fig. 2. Surface tab views in CiteView. Panel A shows context-aware bullet points that explain the cited paper's role in the current citation, panel B visualizes citation-purpose distributions, and panel C lists example citing papers that use the same work, each card revealing the exact quotation when clicked.

alongside the cited text, enabling efficient verification and deeper cross-document sensemaking without disrupting the reader's flow.

3 System Overview

CiteView is an interactive PDF reader that turns static scholarly articles into a citation-centric exploration environment. When a user uploads a paper, the system parses the document layout, detects inline citations, and renders each as a clickable entry point that opens a compact card showing the cited work's metadata and access links (Figure 1). From there, an "Analyze" action invokes a right-hand sidebar organized into four coordinated views, Surface, Compare, Viz, and Results, which together provide layered access to contextual summaries, citation-purpose signals, cross-paper alignments, visual structure, and key findings for each cited paper directly within the original PDF.

3.1 Surface Tab

The Surface tab is the default view that opens when a reader clicks a citation, providing a fast, context-aware explanation of what the cited paper contributes in that specific place in the source document. It is designed not to summarize the entire article, but to answer a narrower question: "Why is this paper being cited here, and what does it add?" To achieve this, it combines the local citation context in the source PDF with selectively extracted content from the cited paper and transforms it into three to four concise, evidence-grounded bullet points that can be read at a glance. (Figure 2)

When a user clicks a citation, the system captures the surrounding context and treats it as a query encoding both the topic and rhetorical role of the citation (e.g., introducing a method, supporting a result). The cited paper is segmented

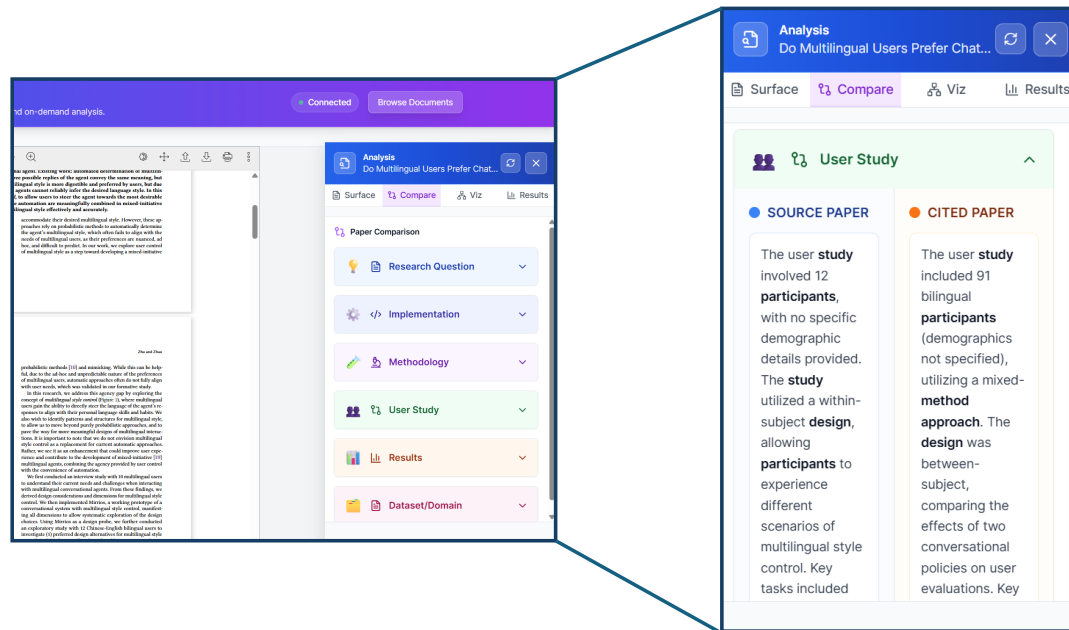


Fig. 3. Compare tab in CiteView. The left panel shows the main PDF reader with the Compare tab selected, and the zoomed view highlights a user-study card comparing the source paper (left column) and the cited paper (right column) along participants, design, and other study details.

into sentences, each scored for relevance using term-weighted similarity that emphasizes shared technical vocabulary. The top-ranked sentences are passed to a large language model configured to produce three to four bullet points, each two sentences long, that: (1) explain the cited work's specific role in the argument, (2) tie that role to the local context, and (3) foreground concrete methods, results, or implications rather than generic summaries.

In the interface, the Surface tab presents these bullets alongside the original citation context, allowing readers to see both the author's sentence and the system's interpretation in one place. Each bullet is initially displayed in a compact form, with controls that allow users to expand it for full text when more detail is needed, supporting both quick scanning and in-depth inspection.(Figure 2.A)

The Surface view also displays citation-purpose distributions showing whether the work is typically used as background, method, or results support, derived from Semantic Scholar's citation-intent labels (Figure 2.B). These aggregated distributions appear as a chart that might reveal, for example, that a work is cited predominantly as methodological support and rarely as critique.

Below the distribution, the tab presents example citations from other papers, each annotated with citation-purpose labels and metadata (title, venue, year). Clicking a card reveals the exact quotation from that source, allowing readers to quickly see how the community uses the cited paper—as background, methodology, or comparative results (Figure 2.C).

3.2 Compare Tab

The Compare tab enables readers to understand how a cited work relates to the current paper across six research dimensions: research question, methodology, implementation, user study, results, and dataset/domain (Figure 3).

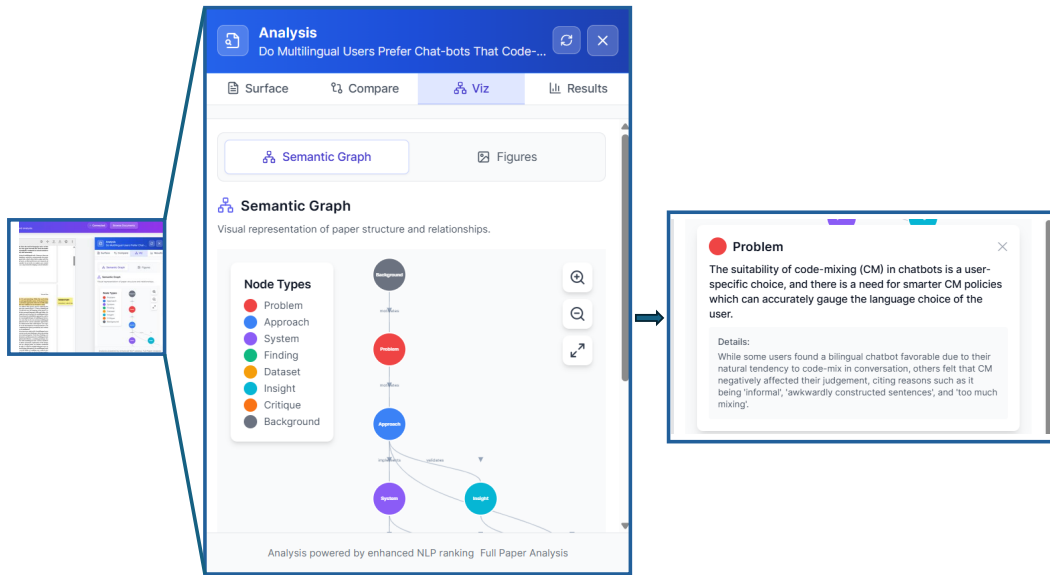


Fig. 4. Semantic Graph view showing the conceptual structure of a cited paper. Clicking on a node (e.g., Problem or Approach) opens a detailed explanation, supporting overview-to-detail exploration without leaving the reading context.

Rather than requiring readers to switch between documents or use external tools, it presents a structured side-by-side comparison that highlights similarities and differences.

When a reader opens the Compare tab, CiteView extracts key content from both papers using a multi-agent system built with the CrewAI framework. A comparison agent synthesizes this content into six paired descriptions, one for the source paper, one for the cited paper, phrased in plain language that foregrounds concrete details such as system components, user study design, or performance metrics. These comparisons appear as collapsible cards in two columns (source left, cited right), allowing readers to quickly scan across dimensions or inspect specific aspects in detail.

This design turns what is usually a slow, manual process of flipping between PDFs into a compact comparative overview that supports higher-level reasoning about the literature, such as identifying complementary methods, spotting opportunities for extension, or detecting when a citation is methodologically misaligned with the work that cites it.

3.3 Viz Tab

The Visualization tab is designed to support rapid structural understanding of cited papers through visual abstractions. It consists of two complementary sub-tabs: Graph and Figure. Together, these views move beyond linear text summaries by exposing the conceptual structure of a paper and its visual evidence, enabling readers to reason about cited work more efficiently within the reading context.

3.3.1 Graph. The Graph view generates a compact semantic knowledge graph capturing the cited paper’s argumentative structure (Figure 4). CiteView extracts and segments the paper’s text, then processes each section with targeted prompts to identify concise statements representing key conceptual units: Problem, Approach, System, Finding, Dataset, Insight,

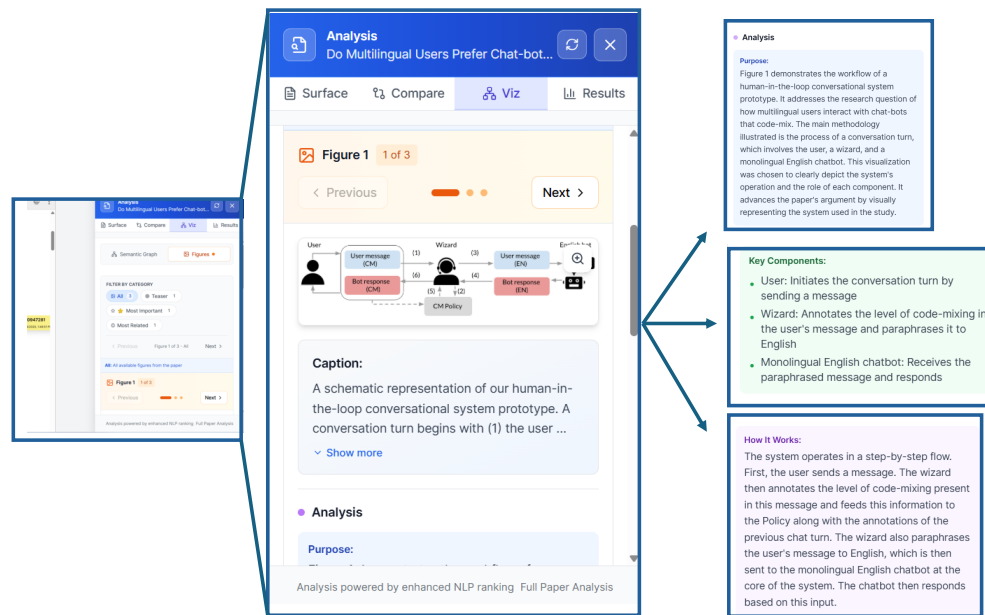


Fig. 5. The Figure view presents a curated set of figures from a cited paper along with structured explanations. Readers can navigate between selected figures (teaser, most important, and most related) and access concise analyses that describe each figure's purpose, key components, and operational flow, enabling rapid interpretation of visual evidence without leaving the reading context.

Background, and Critique. These statements become interactive nodes connected by typed edges (e.g., motivates, implements, produces) that reflect structural relationships. When readers click a node, CiteView displays a focused textual explanation with supporting details, enabling overview-to-detail exploration without linear reading.

3.3.2 Figure. The Figure view surfaces three curated figures from the cited paper: a teaser figure providing high-level overview, the most important figure reflecting the paper's primary contribution, and the most related figure relevant to the current citation context (Figure 6). CiteView extracts figures and captions from the PDF and ranks them using layout heuristics and relevance scoring. Each figure is accompanied by a structured explanation summarizing its purpose, key components, and function, enabling interpretation without consulting the full paper.

3.4 Result Tab

The Results tab supports in-depth understanding of how cited papers report findings (Figure ??). It is organized into three complementary views:

- **Quantitative** presents numerical outcomes as self-contained units highlighting the measured metric, reported value, comparisons, and statistical significance.
- **Qualitative** surfaces themes and user insights as concise claims paired with supporting evidence from the paper.
- **Relation** explains how the cited paper's results connect to the current paper—through validation, comparison, extension, or critique.

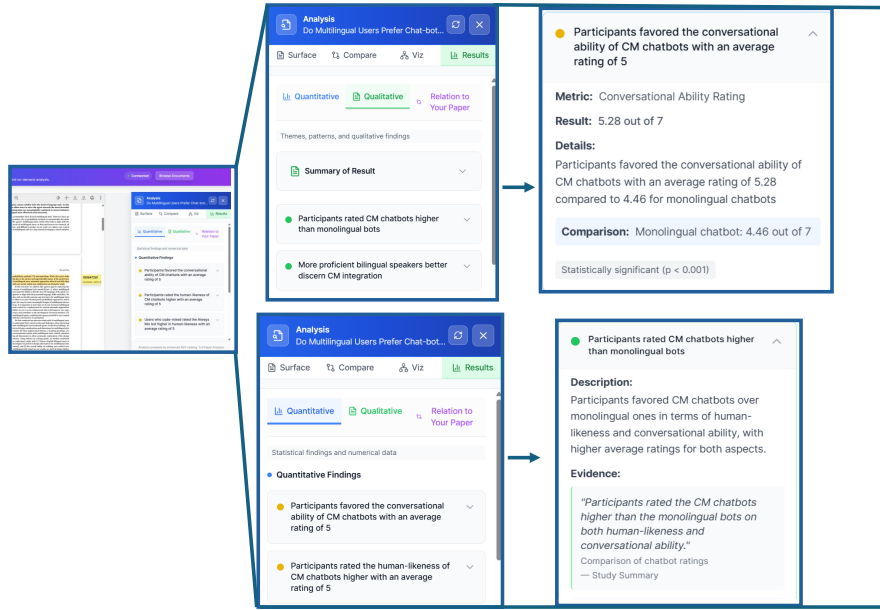


Fig. 6. Results tab showing quantitative, qualitative, and relation views that summarize cited paper findings and explain their relevance to the current paper, supporting efficient interpretation without leaving the reading context.

Content is extracted and analyzed by specialized agents: a Quantitative Analysis Agent for numerical findings, a Qualitative Analysis Agent for themes and patterns, and a Relation Analysis Agent for cross-paper connections.

4 System Implementation

CiteView is implemented as a full-stack web system that integrates structured document processing, coordinate-aware PDF rendering, external scholarly metadata services, and LLM-based synthesis. The implementation emphasizes reproducibility, low-latency interaction, and robustness through hybrid rule-based and learning-based methods. This section details the technical components and algorithms used to transform uploaded PDFs into analysis-ready, citation-aware artifacts.

PDF documents are rendered using **PDF.js** with page-level rasterization and consistent coordinates for interactive overlays. To extract structured bibliographic data, CiteView uses **GROBID** to parse PDFs into TEI XML, providing normalized references with titles, authors, venues, and DOIs. Because GROBID does not expose citation positions, **PyMuPDF** detects citation markers via regular expressions and recovers bounding boxes in page coordinates. Citation highlights are drawn as scaled overlay rectangles, with user interactions mapped to citation identifiers that trigger analysis requests.

Bibliographic entries are enriched using **Semantic Scholar** with prioritized lookup (DOI first, then title-based search). A persistent SQLite cache reduces API load by storing enriched metadata for reuse. When full cited papers are needed, CiteView queries open-access endpoints including Semantic Scholar, Unpaywall, arXiv, and PubMed Central.

To reduce latency and API load, CiteView employs a persistent SQLite cache keyed by normalized identifiers. Cache hits bypass external requests, while cache misses trigger controlled API queries with rate limiting. Enriched metadata is stored and reused across sessions, enabling faster interaction for frequently cited papers.

When deeper analysis requires access to full cited papers, CiteView attempts on-demand PDF acquisition using a multi-source strategy. The system queries open-access endpoints including Semantic Scholar, Unpaywall, arXiv, and PubMed Central. Retrieved files are validated for PDF integrity and cached locally for subsequent use. If no accessible version is found, the system gracefully degrades to metadata-only analysis.

To identify relevant content from cited papers, CiteView treats the citation context as a query and performs targeted retrieval. Candidate sentences are ranked using hybrid scoring: dense semantic similarity (sentence embeddings), sparse lexical similarity (TF-IDF term weighting), structural weighting (prioritizing methods/results sections), and positional heuristics (favoring concise declarative sentences). Top-ranked snippets are passed to the LLM within prompt budget constraints.

Selected evidence snippets are passed to a large language model with citation context and metadata. The model operates exclusively over retrieved evidence, producing grounded summaries returned as structured JSON for consistent rendering.

To minimize hallucination risks, CiteView constrains LLM outputs through evidence-grounding (operating only over retrieved snippets), structured JSON schemas (enforcing probability distributions), external validation (using Semantic Scholar’s curated citation labels), and explicit prompts instructing refusal when evidence is insufficient. Routine classification uses rule-based keyword matching as deterministic fallbacks.

For comparative and results-focused analysis, CiteView employs a multi-agent system built with the CrewAI framework. Specialized agents extract and align content: a Quantitative Analysis Agent identifies numerical findings, a Qualitative Analysis Agent surfaces themes and insights, and a Comparison Agent synthesizes cross-paper relationships.

Figures are extracted using PyMuPDF to identify embedded image objects and captions. Figures are ranked by position, size, and caption informativeness, with the system selecting three representative figures: teaser, most important, and most related to citation context. Semantic graphs are generated by segmenting the cited paper into sections and extracting typed statements (Problem, Approach, System, Finding, Dataset, Insight, Background, Critique) limited to 5-10 nodes. Relationships are inferred through rule-based heuristics (e.g., Problem motivates Approach) combined with semantic similarity scoring.

5 User Study

We conducted a user study to evaluate how CiteView supports citation sensemaking during academic paper reading. The study focused on understanding how the system affects reading flow, citation comprehension, and users’ ability to reason about cited work without leaving the source document.

5.1 Participants

We recruited 12 participants (4 women, 8 men). Ten participants were master’s students, and two were final-year undergraduate students. Most participants were studying computer science, while others came from related fields such as science and management. All participants reported regular experience reading academic papers and using citations to understand prior work.

5.2 Pilot Study

Prior to the main study, we conducted a pilot study with two master's students in computer science to validate the study procedure and identify potential usability issues. Overall, participants responded positively to the system and provided several suggestions for improvement. Based on this feedback, we made minor refinements to the interface and clarified aspects of the study instructions before conducting the full user study. Data from the pilot study were not included in the final analysis. It also helped confirm that the study duration (approximately one hour) was appropriate and that participants could complete the tasks without fatigue.

5.3 Study Design and Procedure

At the beginning of each study session, we introduced CiteView to participants and explained the purpose of the system and the functionality of its different tabs (Surface, Compare, Results, and Visualization). Participants were then given time to explore the system freely and become familiar with the interface and interaction flow. After this familiarization phase, we explained the tasks participants would complete during the study.

Each participant completed three task blocks. To reduce cognitive overload and avoid confusion when interacting with a novel system, the first two blocks focused on a single citation. In these blocks, the interaction was intentionally scaffolded to guide participants through the system's different views.

In the first block, participants were instructed to use the *Surface* and *Compare* tabs to answer four questions, two questions corresponding to each tab. These questions focused on understanding the cited paper's main contribution and how it related to the source paper. In the second block, participants worked with the same citation but were asked to answer four questions using the *Results* and *Visualization* tabs, encouraging deeper engagement with empirical findings and figures from the cited paper.

The third block was designed to assess more naturalistic use of the system. Participants were assigned a different citation and asked to answer four questions without any constraints on which tabs to use. In this block, participants were required to decide for themselves which tabs were most appropriate for finding the necessary information. By this stage, participants had already explored all available tabs in the earlier blocks and were therefore more comfortable navigating the system independently.

This block-based design allowed participants to progressively build familiarity with CiteView while ensuring that all tabs were explored. It also enabled us to observe how participants selected and combined tabs once they had developed an understanding of the system's capabilities.

Following task completion, participants filled out the NASA-TLX and System Usability Scale (SUS) questionnaires and participated in a semi-structured interview reflecting on their experience with CiteView.

5.4 Measures and Data Collection

We collected both quantitative and qualitative data. After completing the tasks, participants completed the NASA-TLX questionnaire to assess perceived workload and the System Usability Scale (SUS) to evaluate overall usability. In addition, we conducted a semi-structured interview to gather participants' subjective impressions, perceived benefits, and limitations of the system.

To better understand interaction patterns, the system also logged user interactions, including the amount of time participants spent in each CiteView tab (Surface, Compare, Results, and Visualization) and the frequency of tab switching during the tasks.

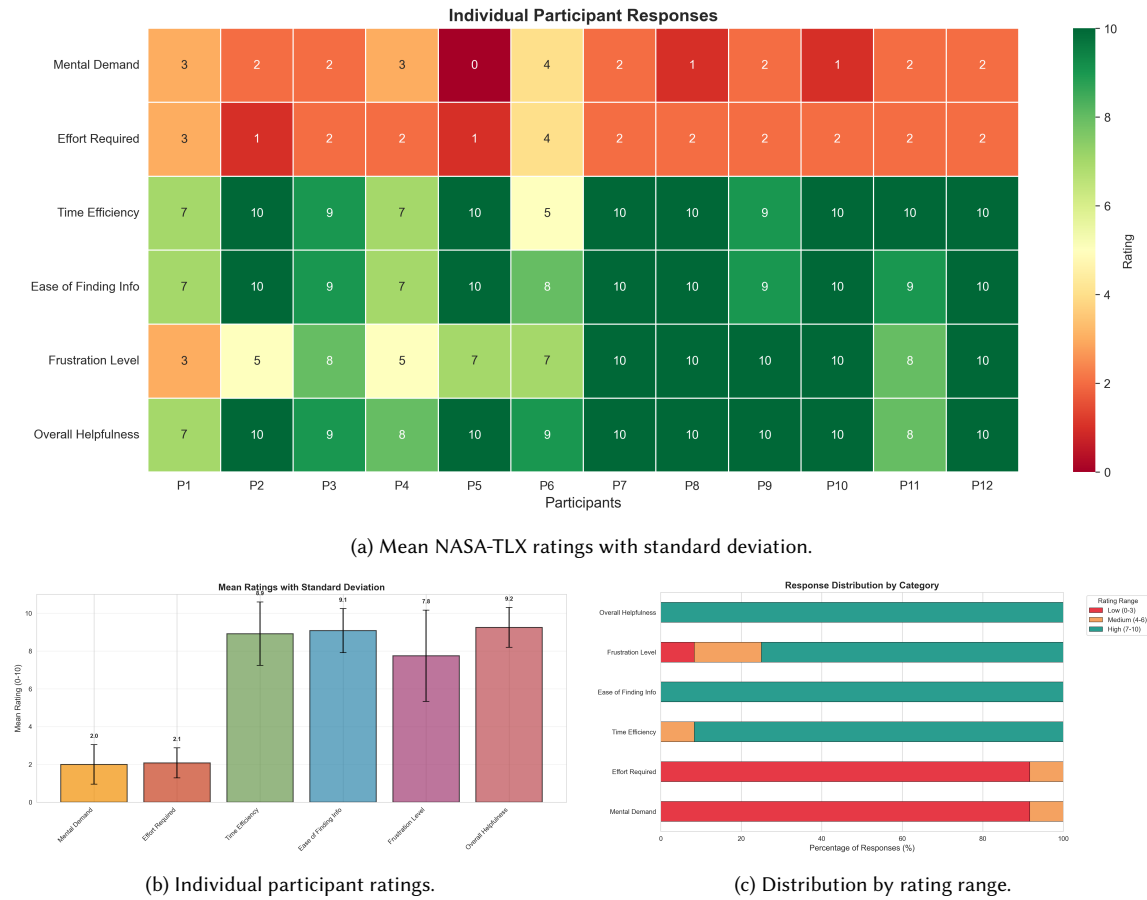


Fig. 7. NASA-TLX workload and perceived task support for CiteView ($n = 12$, 0–10 scale). Lower is better for Mental Demand and Effort Required; higher is better for Time Efficiency, Ease of Finding Information, and Overall Helpfulness. Frustration Level is shown as reported. Panel (A) summarizes central tendency, (B) shows individual variation, and (C) shows rating distributions (Low: 0–3, Medium: 4–6, High: 7–10).

6 Results

We report results from a mixed-methods user study combining interaction logs, standardized questionnaires (NASA-TLX and SUS), and post-task semi-structured interviews. Overall, participants were able to complete citation-focused tasks using CiteView and reported positive experiences with the system. Usability scores were high and perceived workload was generally low, while interview feedback highlighted both the benefits of inline citation exploration and areas for improvement. The following sections detail task performance, questionnaire results, and recurring themes from participant feedback.

6.1 Task Performance

Across all study sessions, participants were generally able to answer the citation-focused questions correctly using CiteView. Most participants answered all questions correctly for the majority of citations, with occasional skipped

or incorrect responses limited to one or two questions per task. Errors and omissions were typically associated with ambiguity in the question wording or uncertainty about which aspect of the cited paper to prioritize, rather than difficulty operating the interface itself. Overall, Across $n = 12$ participants (12 questions each), the mean number of correctly answered questions was 11.75/12 (SD=0.62; Mdn=12; range: 10–12), corresponding to a 97.9% completion rate. Ten participants (83.3%) achieved perfect performance (12/12), and 11/12 participants (91.7%) answered at least 11/12 questions. The small number of missed or skipped questions was typically associated with ambiguity in question wording (e.g., uncertainty about what counted as the “main figure”), rather than difficulty operating the system.

6.2 Usability and Workload

SUS results indicated generally positive usability ($M=71.9$, $SD=13.2$, $n=12$), above the industry benchmark (68) and approaching the “good usability” threshold (73.5). Item-level responses showed clear polarity: participants rated positively worded items high (intention to use, ease of use, feature integration, learnability, confidence) while rating negatively worded items low (unnecessary complexity, need for support, inconsistency, cumbersomeness). This pattern suggests CiteView was experienced as learnable and coherent, with remaining friction related to interface organization rather than basic operability (Figure 8).

NASA-TLX ratings indicated efficient citation exploration with low perceived effort. Participants reported low cognitive and physical burden (Mental Demand: $M=2.0$, $SD=1.0$; Effort Required: $M=2.1$, $SD=0.8$ on 0-10 scale), while giving consistently high ratings for speed and information access (Time Efficiency: $M=8.9$, $SD=1.7$; Ease of Finding Information: $M=9.1$, $SD=1.2$; Overall Helpfulness: $M=9.2$, $SD=1.1$). Frustration scores were more mixed ($M=7.8$, $SD=2.4$), suggesting some participants experienced friction with interface organization or visualization clarity (Figure ??).

6.3 Qualitative Themes

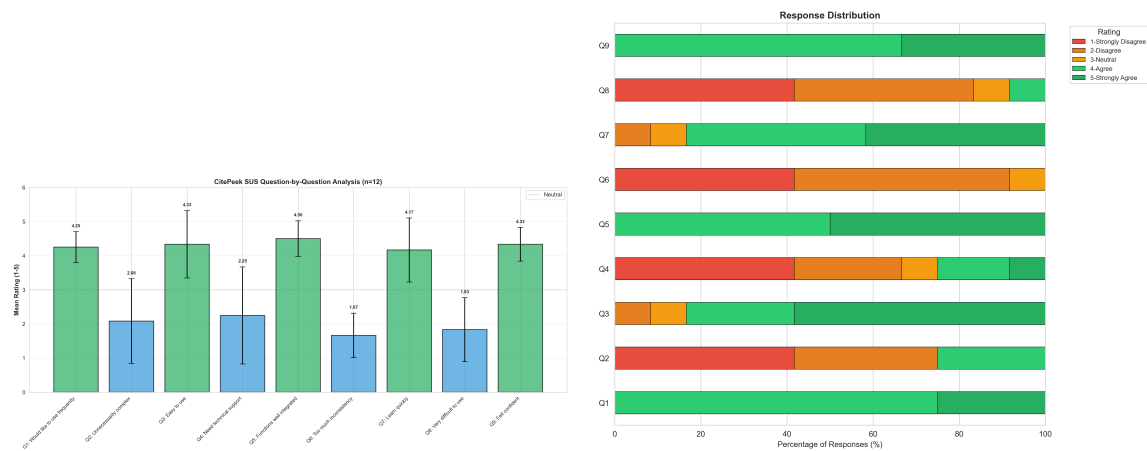
In the post-task interview, we asked participants to reflect on (i) whether CiteView met their expectations and why, (ii) ease of navigation and any confusing elements, (iii) perceived accuracy and usefulness of the information shown in the sidebar, (iv) how the tool compared to their usual citation-following workflow, (v) which features they liked most/least, and (vi) what they would change or add in a future version. Across interviews, three recurring themes emerged: (1) reduced friction in accessing cited papers and contribution-level information, (2) strong preference for comparison- and insight-oriented views, and (3) a desire for a simpler, more consolidated interface with clearer visual explanations.

Theme 1: Reduced friction and faster access to contribution-level information. Participants consistently contrasted CiteView with traditional citation-following workflows, emphasizing speed and eliminated overhead. P1 explained the tool “made the information more readily available and easily accessible compared to looking for that individual PDF,” bringing contributions “instantaneously” and “in a short and concise way.” P4 estimated the efficiency gain was substantial: “It probably reduced it by 10X.” P2 similarly noted that with CiteView, “I can read more papers within a short period of time” compared to the overwhelming process of reading many papers traditionally. P3 highlighted the reduction in authentication and verification steps: without CiteView, they would need to “go on Google, figure out if I had access to that paper,” then “log into my student account, get access through that paper, and then confirm that it’s the paper that’s been cited,” whereas with CiteView, “I just have to click the citation.”

Theme 2: Preference for comparison, insights, and visualization-driven views. When asked about favorite features, participants most often highlighted views that supported interpretation and comparison rather than purely descriptive summaries. P1 stated “the results would be the most favourite because I can instantly understand what the cited paper



(a) Mean SUS ratings (1–5) with standard deviations for each item (n=12). The dashed line indicates the neutral midpoint (3).



(b) Participant-by-item heatmap of SUS responses (P1–P12, Q1–Q9).

(c) Response distributions by SUS item (percent of ratings 1–5 for each question).

Fig. 8. **Item-level SUS results for CiteView (n=12).** Positively worded items show strong agreement, while negatively worded items show disagreement, explaining the overall pattern of favorable usability ratings.

did,” while P2 chose “the comparison, which I really like.” P3 emphasized “the graph and statistics tab because it provides insights really quickly and highlighted the important graphs or figures.” Participants specifically valued the citation

usage visualization, with P6 calling it “a very smart decision” because it shows “what this cited paper was used for the most, or cited in other papers for,” providing “good relevant data that people can use within their actual paper.” This preference aligned with interaction patterns where participants gravitated toward insight- and relationship-oriented views during citation-focused tasks.

Theme 3: Desire for simplification, clearer labeling, and stronger visual explanations. Although participants generally found the interface learnable, several described an initial learning curve and requested reducing verbosity. P1 acknowledged “there’s a little bit of a learning curve initially,” but emphasized it “becomes a lot more” manageable “once you understand what each tab does.” Multiple participants critiqued the Surface tab as too verbose or conceptually unclear. P4 stated they would “delete or modify” Surface because “I don’t understand what surface means,” recommending “just bullet points, bit shorter” instead of verbose paragraphs. P1 suggested to “merge the surface and the results” to achieve “less clutter on the UI,” while P3 wanted the first tab “more condensed” with highlights. P5 summarized the tension: the design was “amazingly designed,” but the main feedback was “getting lost with so many features.”

Participants also requested navigation improvements and better visual explanations. P6 proposed keyword search functionality: “If I could just search within CiteView, it would tell me a paragraph,” and “if there is any figure about it, it gives you an option to go to figures, if there are any results about it, it gives me an option to go to results.” P4 noted visualization clarity issues: “the graph didn’t have a legend that showed what the colours represented, so it was really hard to instantly get what is blue, what is gray.” Other requests included back navigation, clearer figure labeling to avoid ambiguity (P3: explicitly marking “teaser/main figure”), and improved full-screen viewing for visualizations.

Together, quantitative and qualitative results indicate CiteView supports effective citation exploration with low cognitive load, particularly through comparison and visualization features. Key improvement opportunities include consolidating overlapping views, adding keyword search navigation, and improving visual explainability through legends, labels, and interaction affordances.

7 Discussion

Our findings suggest that CiteView reduces citation exploration overhead by making cited-paper contributions, results, and relationships available at the point of reading. Participants completed tasks at high rates (97.9%), reported positive usability (SUS: $M=71.9$), and described substantial time savings, with one participant estimating efficiency gains of “10X.”

NASA-TLX responses revealed low mental demand ($M=2.0$) and effort ($M=2.1$) alongside high time efficiency ($M=8.9$) and ease of finding information ($M=9.1$). Interview feedback explains this pattern: participants repeatedly emphasized one-click access to contribution-level summaries as eliminating common barriers—searching, authentication, and context switching. Together, these results suggest that integrating contribution-oriented views directly into the reading workflow can support faster sensemaking without substantially increasing cognitive load.

A consistent qualitative theme was participants’ preference for insight-oriented tabs (Results, Compare, Visualization) over purely descriptive overviews (Surface). This indicates users value *actionable* information—what the cited paper contributed, how it is used, and what evidence it provides—more than narrative summaries. This suggests condensing overview content or integrating it into insight views to reduce redundancy.

Although usability ratings were positive, remaining friction stemmed from interface organization rather than basic operability. Requests to merge tabs, add keyword search, and improve legends point to a key tradeoff: powerful views

increase capability but may reduce clarity. Future iterations should emphasize progressive disclosure, clearer naming, and stronger visual scaffolding.

Our results suggest several design principles for citation exploration tools: (1) *contribution-first access* through fast summaries that reduce navigation overhead, (2) *insight-driven views* prioritizing results and comparisons over descriptions, (3) *progressive disclosure* to avoid overwhelming users, (4) *self-explanatory visualizations* with clear legends and labels, and (5) *in-tool retrieval support* such as keyword search to align with question-driven workflows.

8 Limitations and Future Work

Our study have several practical limitations. First, CiteView depends on access to the full text of cited papers; however, not all references are openly available due to paywalls, missing PDFs, or incomplete metadata. As a result, some citations cannot be expanded into summaries, comparisons, or figure retrieval, which limits coverage and can introduce inconsistency across papers. Second, document quality and formatting remain a constraint: older or non-standard PDF layouts (e.g., scanned documents, uncommon templates, or poorly structured text layers) may reduce extraction reliability and may not be fully supported by the current pipeline, affecting citation detection, section parsing, and figure linking.

These limitations also motivate several directions for future work. To better support diverse reading contexts and accessibility needs, we plan to explore audio-based interaction (e.g., narrated citation summaries or voice-based playback of key contributions). We also aim to make citation exploration more interactive by adding keyword search and “jump-to” navigation across tabs, enabling users to quickly locate methods, results, or figures related to their question. Another direction is integrating an AI chat interface so users can ask targeted questions about a cited paper (or about how a citation is used in context) and receive grounded answers based on the retrieved content. Finally, consistent with participants’ requests for simplification, we plan to merge overlapping tabs and streamline the information architecture, keeping advanced views available through progressive disclosure while reducing perceived complexity for first-time users.

9 Conclusion

We presented CiteView, an interactive citation exploration tool that brings contribution-level context from referenced papers directly into the reading workflow. CiteView supports question-driven citation use by combining lightweight summaries with comparison, and relationship-oriented views, allowing readers to quickly understand what a cited work contributes and how it connects to the current paper without repeatedly switching tabs, searching externally, or navigating paywalls.

In a mixed-methods user study ($n = 12$), participants reported generally positive usability (SUS: $M = 71.9$, $SD = 13.2$) and low perceived workload on key effort-related dimensions, alongside high perceived time efficiency, ease of finding information, and overall helpfulness. Interview feedback further highlighted that CiteView reduced friction in accessing cited work, and that participants especially valued insight-driven views (e.g., results, comparison, and visualizations) for quickly answering citation-focused questions. At the same time, participants emphasized opportunities to simplify the interface, improve visual explainability, and better support in-tool navigation.

Overall, our findings suggest that citation interfaces should prioritize contribution-first access and insight-oriented representations while using progressive disclosure to balance power and learnability. We believe CiteView takes a step toward more fluid, question-driven scholarly reading by shifting citation exploration from a disruptive detour into an integrated, interpretable part of reading.

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